



ENVE | GEOE 121: Computational Methods

Jaber Taheri-Shakib

Research associate

Civil and Environmental Engineering

University of Waterloo | McMaster University



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❑ Opening Discussion

- 1) Why are computational methods, coding, and programming important in engineering?
- 2) How can these skills help you solve real engineering problems?
- 3) How can computational skills help you in your future career or job search?
- 4) Which do you prefer: visual/practical work, programming, or a combination of both?



What I Hope You Take From This Course:

- ✓ How to use computational tools to **analyze data**.
- ✓ How to break engineering problems into **clear steps**.
- ✓ How to use MATLAB to **calculate, visualize, and interpret results**.
- ✓ How to check whether your results are **reasonable and meaningful**.
- ✓ How computational methods help us understand and solve **real engineering problems**

The Teaching Team



Jaber Taheri-Shakib

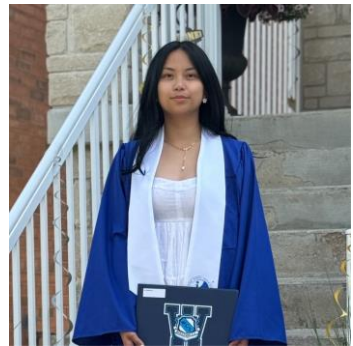
Research Associate

*Structures, Construction, Image Analysis, Material Science,
Corrosion, Flow in Porous Media*

Email: Jtaheris@uwaterloo.ca

Office Hours: Tuesday / Thursday from 1:30–2:30 PM

Location: E2 – 1782



Adrienne Co

Email: a5co@uwaterloo.ca

Office Hours: 9:30 AM to 5:30 PM, Monday through Friday, apart from the allotted lab time.

Location: the First-Year Engineering Office located in E2-1786



Tianyi Zhang

Email: t532zhang@uwaterloo.ca

Office Hours: Wednesday 2.30-3.30

Location: E2-2344B

Course Outline

Lecture Topics

1. MatLab Basics
2. Arrays
3. Intro to Data Handling
4. Decisions
5. Loops
6. Functions
7. Symbolic Computation

Fundamental Skills

Core Programming

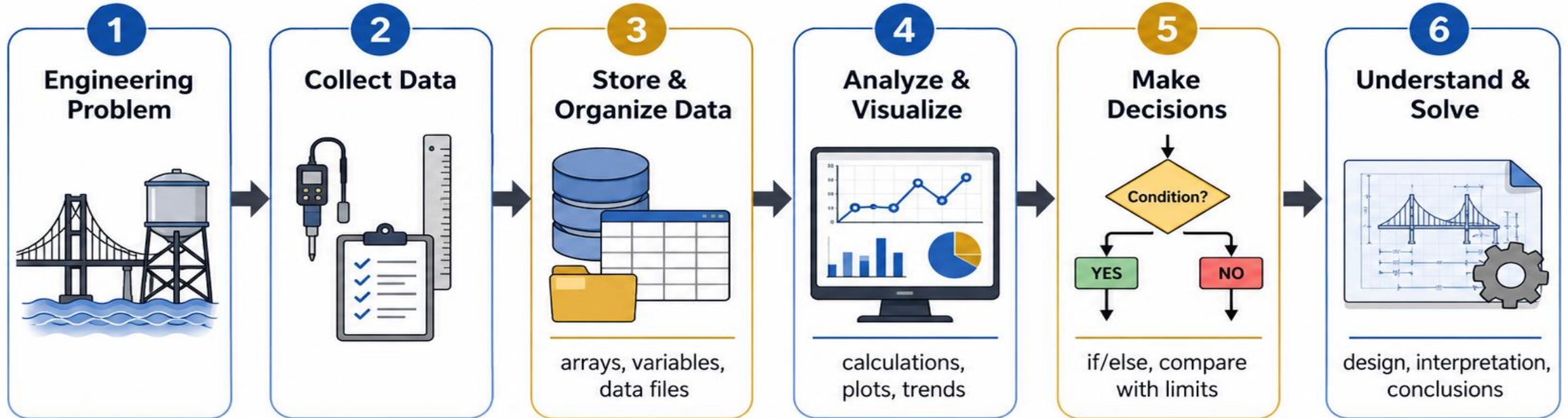
Data Manipulation

Mathematical Analysis

Core Outcome

**Tools for
Engineering
Design
and Analysis**

How Computational Methods Help in Engineering



Computational methods help us organize data, analyze results, and make engineering decisions.

Example: River Water Quality Analysis After Rainfall

A specific environmental engineering example using computational methods

1 Problem

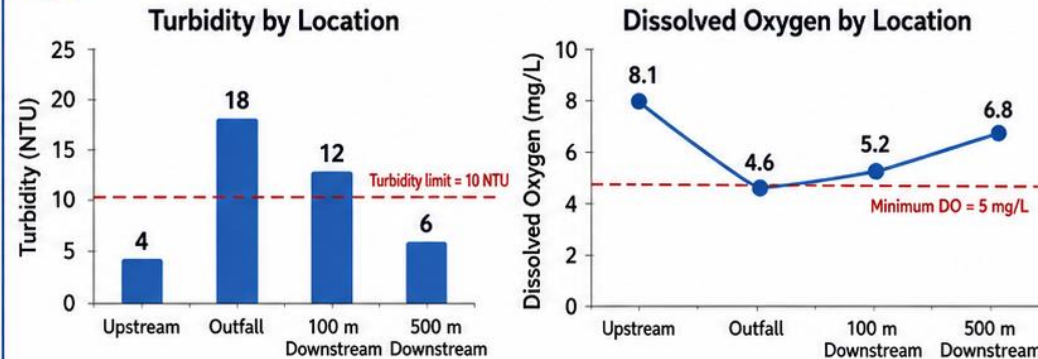


After a rain event, we want to know whether runoff from an outfall is reducing river water quality downstream.

2 Sample Data

Location	Turbidity (NTU)	Dissolved Oxygen (mg/L)
Upstream	4	8.1
Outfall	18	4.6
100 m Downstream	12	5.2
500 m Downstream	6	6.8

3 Analysis



From data to engineering decision

Example: Slope Stability Analysis After Rainfall

A specific geotechnical engineering example using computational methods

1 Problem

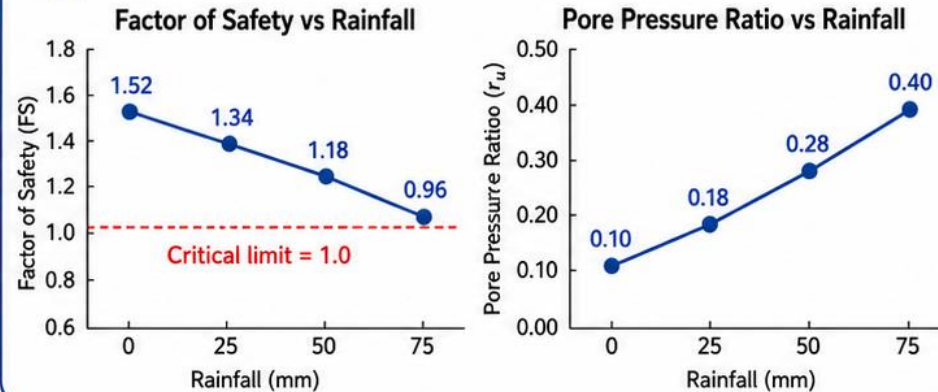


After heavy rainfall, we want to know whether the slope is becoming unsafe. Rainwater increases pore water pressure and can reduce slope stability.

2 Sample Data

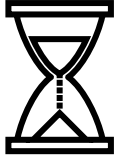
Rainfall (mm)	Pore Pressure Ratio r_u	Shear Strength (kPa)	Factor of Safety
0	0.10	42	1.52
25	0.18	39	1.34
50	0.28	35	1.18
75	0.40	30	0.96

3 Analysis



From data to geotechnical engineering decision

Class Structure

Monday	Tuesday	Wednesday	Thursday	Friday
	Class (11:30 AM) May 14 - Jun 19, Jun 29 - Aug 5 	LAB (3:30 PM) May 11 - Jun 19 Jun 29 - Aug 5  E2 1792	Class (11:30 AM) May 14 - Jun 19, Jun 29 - Aug 5 	LAB (12:30 PM) Jun 26  E2 1792
		HW Due (11 pm) 	Class (04:00 PM) May 14, May 28, Jun 11, Jul 9 	HW Due (11 pm) 

- Homework assignments should be completed individually.
- Labs are to be completed with your assigned partner.
- Assignment questions will be submitted through MatLab Autograder, CrowdMark, and LEARN.

Your output is the proof that your code works. It needs to match the code you've submitted!

Late Submissions and Extensions

Late Submissions

- 24 hours late -> -25%
- 48 hours late -> -50%
- >48 hours late -> -100%

Approved Absences

Any deadlines missed are discounted and reweighted toward the final.

How Can I Earn a 24 hr. Extension?

1. Come to office hours
2. Ask a question about:
 - a. Whether your approach to the problem is reasonable.
 - b. How to approach coding a particular aspect of the problem.
 - c. Suggestions for dealing with an error or a bug.
 - d. Clarification on a course topic.
3. Tell the teaching team you would like an extension

⚠: Unless there are significant extenuating circumstances, extensions can not be claimed via email!

To Those Who Have Coded Before...

And Everyone Who's About to Get Excited About Coding This Term

“Homework assignments should be completed using only the coding structures learned in class prior to the due date of the assignment.”

What This Means For You:

- Until we have had a formal unit on a particular coding structure in class (e.g. functions, file I/O, plotting, decisions, loops, etc.) these structures are prohibited in your assignments. Solutions using these structures will be marked down.
- Once we've covered a particular coding structure in class, you are free to use that structure in creative ways not discussed in lecture (e.g. researching new functions, implementing interesting loops, etc.). You will be graded on these implementations as if they had been taught in class.

Getting Help on Assignments

- You can discuss programming concepts with students in the course or other knowledgeable people. *Acknowledge their help in the comments on your code.*
- You can search for help online through tutorials and GenAI (e.g. StackOverflow, ChatGPT). *Acknowledge these sources in the comments on your code.*
- You cannot copy, read, or provide code to others in this course regardless of the medium (online or in-person). This is considered an academic offence under Policy 71.
- You cannot provide course materials (lecture notes, assignment questions, etc.) to an external website (e.g. StackOverflow, ChatGPT). This is considered an academic offence under Policy 71.
- You can provide assignment questions and lecture examples to the University's version of Co-Pilot to help understand concepts and learn the material. This GenAI software provides some guarantees as to data privacy and security.

Using GenAI

“If you skip the pain of learning, the exam will supply it.”

Can I use GenAI in this course?

Yes! You may use GenAI to support your learning on homework assignments.

What about labs?

Labs are your chance to collaborate with your peers and get support from the TAs. I expect these assignments to be completed independent of GenAI.

Okay... but how do I use it so that I'm prepared for the tests?

DO NOT ask GenAI to solve your homework problems for you. It can do it, but then you won't be able to.

DO ask these tools *why*, *how*, and *what if*?

Digital Communication - Piazza

When to use it

- You are struggling on an assignment and you need some advice.
- You found out something cool about MatLab that you want to share.

Engagement Credit

- You have the opportunity to earn up to 4 bonus points for this course!
- For every month that you ask insightful questions or provide useful answers/information, you may receive a bonus point for the course.

Rules of the Road

- **Be respectful** or your peers and the teaching team.
- Check if the same question has already been asked.
- **Do not post coded solutions** for the homework problems, in full or in part. This undermines the assignment and is considered academic dishonesty.
- Inappropriate posts and comments will be immediately deleted and closed without notice.

Digital Communication - Email

When to use it

- You have a personal question and need advice from the Professor.
- You need support for accommodations regarding an approved absence.

Rules of the Road

- Include the phrase “ENVE | GEOE 121” in the title of your message.
- Dr. Taheri will respond to your messages as promptly as possible, but will not respond outside of the hours of 8am-5pm ET.

Project



Students:

- ✓ 97 Environmental Engineering students
- ✓ 23 Geotechnical Engineering students

Group Format:

- ✓ Each group: **6 students**
- ✓ Presentation time: **12 minutes**
- ✓ Each student: **about 2 minutes**

Next Steps:

- ❖ Find and identify your group today.
- ❖ By next class, today, submit your **group name** and the **names of all group members**.

More details about the project topic, format, and grading rubric will be provided later.

Project



❑ Individual Introduction in Each Group

Each member should briefly answer:

1. What is your name?
2. How did your group choose each other?
3. What is your opinion about your group?
4. What is your role or possible contribution?
5. What is your strongest point that can help the group?

Please be clear, respectful, and honest.

Project



Project Evaluation Focus:

- ✓ Clear problem statement
- ✓ Logical analysis and solution
- ✓ Quality of content and engineering understanding
- ✓ Slide design and organization
- ✓ Time management
- ✓ Teamwork and equal contribution

Note: Speaking fluency is not the main focus. The priority is your understanding, analysis, and ability to clearly explain the problem and solution.

Project



☐ Feedback should focus on:

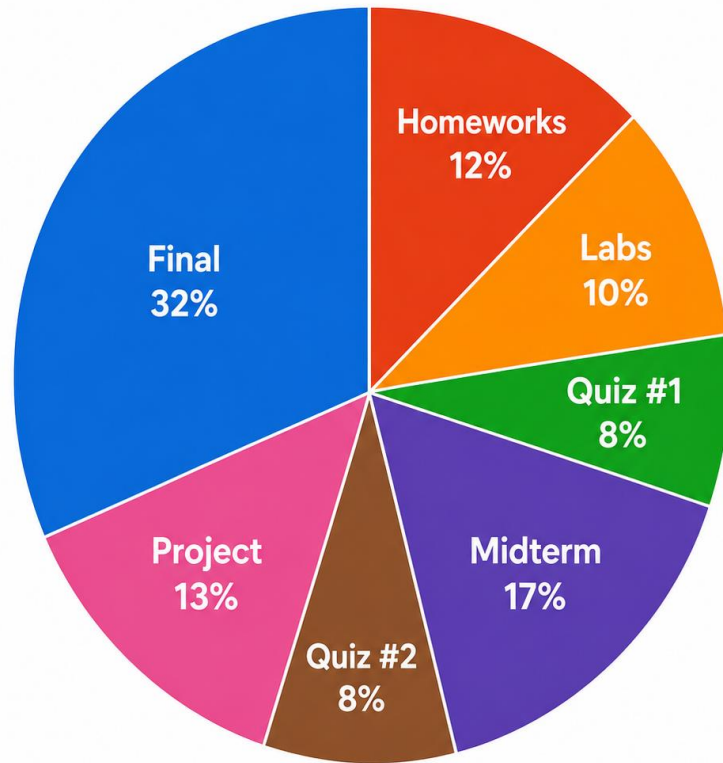
- ✓ Problem statement
- ✓ Analysis and solution
- ✓ Slide design and organization
- ✓ Time management
- ✓ Clarity of explanation

Assessments

- We will hold Two Quizzes and Two Exams:
 - ❖ Quiz 1 (Topics 1-2) – Tuesday, June 16, 2026 - 4:00 PM - 5:20 PM - UW STC 0060
 - ❖ Midterm (Topics 1-3) – June 26, 2026 - 12:30 PM to 2:20 PM - E2-1792 and CPH 1346
 - ❖ Quiz 2 (Topics 1-5) – Thursday, July 23, 2026 - 4:00 PM - 5:20 PM - UW STC 0060
 - ❖ Final (Topics 1-7) – Final Exam Period
- Assessments will include both programming questions and short answer/multiple choice concept questions.
- Assessments are handwritten, closed book.

Marking Scheme

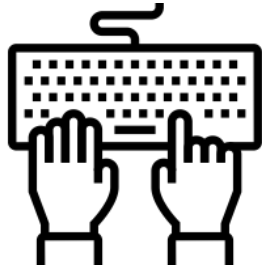
Standard Weighting Breakdown



Activity: extra, up to 8%

Deadlines that are missed due to approved absences (Illness Self-Declaration, General Self-Declared Absence, etc.) will be discounted and the corresponding weight will be reassigned to the final.

How to Do Well in This Course



Keep Practicing

- Rework examples shown in class on your own MATLAB Terminal.
- Work in MatLab directly, rather than in MATLAB Autograder or text editors.
- Connect with the teaching team and other class members to work through difficult concepts.



Take Care of Yourself

- Take time out to breathe.
- Get in touch with [Engineering Wellness](#) and [Campus Wellness](#) to support your mental and physical health.
- Get connected with tutors and other success strategies through the [Student Success Office](#).
- Please make time for exercise

Recommended References

Course Notes

- Lecture materials for the upcoming week are posted on LEARN.

Course Textbooks

- Stormy Attaway, *MatLab – A Practical Introduction to Programming and Problem Solving*, 5th edition, 2017.
 - Available online through UW Library Services
- Holly Moore, *MatLab for Engineers*, 5th edition, 2017.

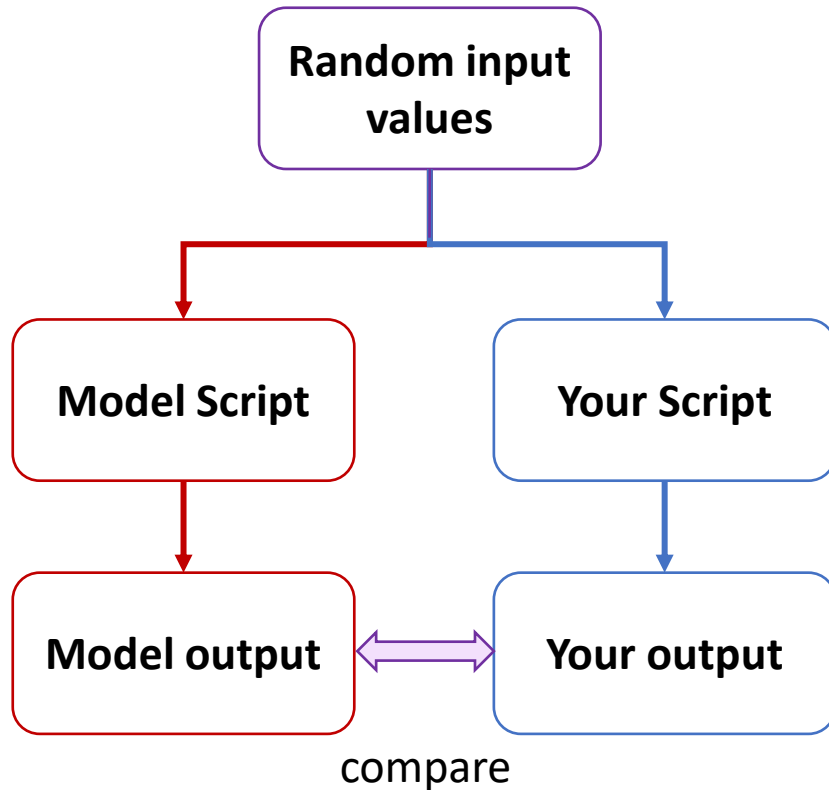


Homework – Due Monday

- Sign up for MatLab Grader
- In LEARN, in the *Introductory Logistics Module*
 - Register for Piazza
 - Complete the Academic Integrity Quiz on LEARN



MatLab Grader



: The random number generator is to ensure that your code is a program, and not a calculator script (no hard-coding!).

MATLAB® Grader™ is a browser-based environment for creating and sharing MATLAB coding problems and assessments. It's an auto-grading system.

An instructor designs several testers (assessments) to check your script if

- Correct outputs are generated from the given random inputs;
- Variables are properly defined;
- Keywords (e.g., built-in function) are present or absent.

Now You Know

- Why you're here, and what you should get out of this course.
- What to expect from the term, regarding assignments, grading, and communication.
- How to succeed in ENVE|GEOE 121: Computational Methods.

Your Grades Matter

